Ben M. Andrew

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Education

University of Manchester, PhD in Computer Science 2024–Present

- Research focused on formal verification of degraded systems.
- Supervised by Dr. Marie Farrell, Prof. Louise Dennis, and Prof. Michael Fisher.

University of Cambridge, BA (Hons) in Computer Science 2019–2022

- Grade II.1 (70%).
- Dissertation titled *Delay-Tolerant Link-State Routing*. Implemented and empirically evaluated a failure-tolerant routing protocol in C to run on real Unix-based routers, comparing it against traditional protocols. Supervised by Dr. Jackson Woodruff.

Publications

- **Counterexample-Guided Interval Weakening** 2026
  Andrew, B. M., Dennis, L. A., Fisher, M., Farrell, M., ABZ 2026, paper
- **Weakening Goals in Logical Specifications** 2025
  Andrew, B. M., ABZ 2025, paper

Industry Experience

Software Engineer, Tarides – Paris, France 2022–2024

- Built open-source CI infrastructure for the OCaml ecosystem, distributing dependency solving, compilation, and testing across a broad matrix of operating systems, CPU architectures, and compiler versions.
- Tested tens of thousands of packages across the Opam Repository, OCaml's central package universe.
- Exposed concrete bugs in the OCaml 5 multicore runtime through systematic stress-testing, reporting and tracking them to resolution upstream.

Embedded Systems Intern, Tarides – Paris, France 2021

- Cross-compiled the OCaml bytecode runtime to an ARM-based microcontroller running a real-time operating system, as part of an R&D project.

Computer Graphics Research Intern, University of Cambridge 2020

- Developed and implemented a visual model of ageing vision in virtual reality with Unity, as part of the Graphics and Displays Research Group.

Teaching Experience

University of Cambridge, Supervisor (TA equiv.) 2022–Present

- Leading small group supervisions for undergraduate courses in computer science, including topics such as logic, formal verification, and semantics of programming languages.

University of Manchester, Graduate Teaching Assistant 2024–2026

- Assisted in teaching graduate and undergraduate courses in computer science, including topics such as software engineering, logic, and formal verification.

Skills

Topics: Research, Computer Science, Mathematics, Systems Programming, Formal Verification, Software Engineering

Programming: AI-assisted programming, C, C++, OCaml, Python, Rust, HTML, CSS, Bash, Git, Docker

Languages: English, French